Control work 2 · Rational expressions

Multiplication, division and long chains, then the work-on-mistakes lesson that turns the paper into teaching.

LESSON 23–24 | 2 × 40 MIN | ALGEBRA 8, §§5–6 | STAGE 9 · UNITS 2, 8

LESSON CLOCK

2' | 36' | 2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess multiplication, division, powers and combined transformations.
- Identify which of the four standard errors each lost mark belongs to.
- Re-solve every lost task in full.

TEXTBOOK

National	Revision of §§5–6
Cambridge	Learner's Book pp. 21–54, 161–189
Workbook	Workbook Units 2, 8

01 Explanation

Lesson 23 — the paper (40 minutes)

Two variants of seven tasks. The balance of the paper:

- tasks 1–2 · multiplication and division of fractions (§5)
- task 3 · a power of a fraction (§5)
- tasks 4–5 · a combined expression with brackets (§6)
- task 6 · a compound fraction (§6)
- task 7 · prove an identity (§6)

Marking: 2 marks per task, 14 in total. Task 7 earns nothing for substituting a single value — an identity has to be proved for all permissible values.

Lesson 24 — work on mistakes (40 minutes)

The errors in this paper fall into four groups, and they are not the same four as in Control work 1:

WHAT GOES WRONG IN §§5–6

1. **Not turning the divisor over** — multiplying by $\frac{C}{D}$ instead of $\frac{D}{C}$.

2. **Expanding before cancelling**, producing a page of algebra and then an arithmetic slip.
3. **The index not reaching the coefficient** — $(\frac{3x}{2})^2$ written as $\frac{3x^2}{2}$.
4. **Treating an identity as an equation** and moving terms across the equals sign.

The Hard set below is seven pieces of wrong working drawn from these four. Diagnose, then repair.

02 Worked examples

EXAMPLE 1 **Find and repair the error:** $\frac{x}{3} : \frac{x}{6} = \frac{x^2}{18}$

1. The divisor was multiplied, not inverted.

   This is error 1.

2. $\frac{x}{3} \cdot \frac{6}{x}$

   Turn the second fraction over.

3. $\frac{6}{3} = 2$

   The x cancels.

ANSWER

$2, \ x \neq 0$

EXAMPLE 2 **Find and repair the error: $\left(\frac{3x}{2}\right)^2 = \frac{3x^2}{2}$**

1. The index must reach every factor, above and below.

   This is error 3.

2. $\left(\frac{3x}{2}\right)^2 = \frac{(3x)^2}{2^2}$

   Square the whole numerator and the whole denominator.

3. $\frac{9x^2}{4}$

   $3^2 = 9$ and $2^2 = 4$.

ANSWER

$\frac{9x^2}{4}$

03 Interactive model

Lesson 24: show a piece of wrong working, take a vote on which of the four errors it is, then reveal.

Diagnose the error

Claimed: $\dfrac{a}{b} : \dfrac{c}{d} = \dfrac{ac}{bd}$

1. Dividing by a fraction means multiplying by its reciprocal.

2. $\dfrac{a}{b} \cdot \dfrac{d}{c}$

The second fraction turns over.

3. $\dfrac{ad}{bc}$

Compare with the claim — the c and d are the wrong way round.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Reciprocal	Teskari	Обратный
Coefficient	Koeffitsiyent	Коэффициент
Expand	Ochish (yoyish)	Раскрыть
Diagnose	Aniqlash	Определить (диагностировать)
Justify	Asoslash	Обосновать
Identity	Ayniyat	Тождество

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\frac{x}{3} : \frac{x}{6}$ equals:

- $\frac{x^2}{18}$
- 2
- $\frac{1}{2}$
- $\frac{x}{18}$

$\left(\frac{2a}{5}\right)^3$ equals:

- $\frac{8a^3}{125}$
- $\frac{2a^3}{125}$
- $\frac{6a^3}{15}$
- $\frac{8a^3}{5}$

The fastest first move on $\frac{x^2 - 4}{x + 2} \cdot \frac{x + 2}{x - 2}$ is:

- multiply out the numerators
- factorise and cancel
- find a common denominator
- substitute a value of x

To prove an identity you may:

- multiply both sides by the LCD
- move a term across the equals sign
- transform one side into the other
- test one value of x

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. ***Variant 1, task 1.*** $\frac{3}{x} \cdot \frac{x^2}{9}$

ANSWER $\frac{x}{3}$

2. ***Task 2.*** $\frac{a}{4} : \frac{a}{8}$

ANSWER 2

3. ***Task 3.*** $(\frac{2x}{3})^2$

ANSWER $\frac{4x^2}{9}$

4. ***Task 4.*** $\frac{x^2 - 1}{x} \cdot \frac{x}{x + 1}$

ANSWER $x - 1$

5. ***Task 5.*** $1 + \frac{2}{x}$

ANSWER $\frac{x + 2}{x}$

6. ***Task 6.*** $\frac{1 + \frac{1}{x}}{x + 1}$

ANSWER $\frac{1}{x}, x \neq 0, -1$

7. ***Task 7.*** *Prove* $\frac{1}{x} - \frac{1}{x + 1} = \frac{1}{x(x + 1)}$

ANSWER LCD $x(x + 1)$; numerator $(x + 1) - x = 1$.

Medium · the standard question

7 PROBLEMS

1. ***Variant 2, task 1.*** $\frac{5}{a} \cdot \frac{a^2}{10}$

   ANSWER $\frac{a}{2}$

2. ***Task 2.*** $\frac{x}{6} : \frac{x}{3}$

   ANSWER $\frac{1}{2}$

3. ***Task 3.*** $(\frac{3a}{4b})^2$

   ANSWER $\frac{9a^2}{16b^2}$

4. ***Task 4.*** $\frac{x^2 - 9}{x + 3} \cdot \frac{x}{x - 3}$

   ANSWER $x, x \neq \pm 3$

5. ***Task 5.*** $\frac{a}{a + b} + \frac{b}{a + b}$

   ANSWER $1, a \neq -b$

6. ***Task 6.*** $\frac{\frac{1}{x} - \frac{1}{y}}{\frac{1}{x} + \frac{1}{y}}$

   ANSWER $\frac{y - x}{y + x}$

7. ***Task 7.*** *Prove* $\frac{a}{a - b} + \frac{b}{b - a} = 1$

   ANSWER $b - a = -(a - b)$, so the sum is $\frac{a - b}{a - b} = 1$.

Hard · stretch and reason

7 PROBLEMS

1. *Find the error:* $\dfrac{a}{b} : \dfrac{c}{d} = \dfrac{ac}{bd}$

ANSWER The divisor was not inverted. Correct: $\dfrac{ad}{bc}$.

2. *Find the error:* $(\dfrac{3x}{2})^2 = \dfrac{3x^2}{2}$

ANSWER The index missed the 3 and the 2. Correct: $\dfrac{9x^2}{4}$.

3. *Find the error:* $\dfrac{x^2 - 4}{x + 2} \cdot \dfrac{x + 2}{x - 2} = x^2 - 4$

ANSWER Nothing was cancelled. Correct: 1, for $x \neq \pm 2$.

4. *Find the error: to prove* $\dfrac{1}{x - 1} - \dfrac{1}{x + 1} = \dfrac{2}{x^2 - 1}$, *"multiply both sides by* $x^2 - 1$*"*

ANSWER An identity is proved by transforming one side, not by operating on both.

5. *Find the error:* $\dfrac{x}{2} : 2 = x$

ANSWER Dividing by 2 means multiplying by $\dfrac{1}{2}$. Correct: $\dfrac{x}{4}$.

6. *Find the error:* $(\dfrac{1}{x} + \dfrac{1}{y})^{-1} = x + y$

ANSWER The reciprocal of a sum is not the sum of reciprocals. Correct: $\dfrac{xy}{x + y}$.

7. *Find the error:* $\dfrac{x^2}{x} : x = x$

ANSWER $\dfrac{x^2}{x} = x$, and $x : x = 1$. Correct answer 1, for $x \neq 0$.

07 Homework

Homework after the work-on-mistakes lesson

Re-solve every task you lost marks on, then these three.

1. $\dfrac{x^2 - 16}{x} : \dfrac{x + 4}{x^2}$
2. $\left(\dfrac{1}{a} - \dfrac{1}{b}\right) : \dfrac{b - a}{ab}$
3. Write out the four errors listed in this lesson with one example of each, in your own words.